Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/29/2020
PI Name:	Roberts, Donna M.D.		
Project Title:	Safety and Efficacy of an Accelerated Protocol of Intermittent Theta Burst Transcranial Magnetic Stimulation (TMS) to Enhance Performance and Promote Resilience in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	George, Mark M.D. (Medical University of South Carolina) McTeague, Lisa Ph.D. (South Carolina Education Association)		
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Task Description:	Given the extreme conditions and stressful situations that will be encountered on exploration class missions, crews must demonstrate peak performance and resilience. Beyond medications, nutrition, and exercise, non-invasive brain stimulation is a method for maintaining and optimizing performance. Here we will develop and establish the efficacy of accelerated repetitive transcranial magnetic stimulation (rTMS) to left dorsolateral prefrontal cortex (dlPFC) as a non-pharmacological method for enhancing cognitive performance and resilience in high-performing healthy adults of astronaut age. First, we will establish an optimal dose of accelerated rTMS for enhancing neurocognitive performance and resilience that maximizes efficacy while minimizing burden. We will then examine the efficacy and safety of accelerated rTMS to left dlPFC for enhancing cognitive performance and resilience. Finally, we will evaluate the biological efficacy of TMS in a microgravity environment (parabolic flight).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Transcranial magnetic stimulation (TMS) is a medical device that can safely and non-invasively stimulate the brains of awake individuals to induce neuroplasticity and is currently under investigation for numerous applications in behavioral health and performance. We aim to optimize TMS to improve neurocognitive performance and enhance resilience to stress in healthy, astronaut-like subjects for use on future long-duration missions. In addition, we have demonstrated an altered response to TMS during a microgravity analog (parabolic flight). Understanding the response of the brain to acutely altered gravity will facilitate investigators in designing TMS treatment protocols for use on future spaceflight missions. These results could also contribute to the understanding of altered inflight responses reported for psychotropic medications by astronauts and may be useful for physicians and pharmacists when prescribing these medications. Earth Noninvasive brain stimulation is a novel method for enhancing performance across multiple cognitive domains and will be applicable to terrestrial patients as well as space travelers. Furthermore, we developed and validated custom TMS helmets that allowed us to administer TMS in zero gravity with minimal training and high individual reliability. This TMS helmet method allows for reliable administration of TMS outside the laboratory setting with minimal training and may be useful in patient populations for clinical purposes as well.
Task Progress:	1. Given the extreme conditions and stressful situations that will be encountered on exploration class missions, crews must demonstrate peak performance and resilience. Beyond medications, nutrition, and exercise, non-invasive brain stimulation is a method for maintaining and optimizing performance. Here we will develop and establish the efficacy of accelerated repetitive transcranial magnetic stimulation (rTMS) to left dorsolateral prefrontal cortex (dlPFC) as a non-pharmacological method for enhancing cognitive performance and resilience in high-performing healthy adults of astronaut age. First, we will establish an optimal dose of accelerated rTMS for enhancing neurocognitive performance and resilience that maximizes efficacy while minimizing burden (Specific Aim#1). We will then examine the efficacy and safety of accelerated rTMS to left dlPFC for enhancing cognitive performance and resilience (Specific Aim#2). Finally, we will evaluate the biological efficacy of TMS in a microgravity environment (parabolic flight) (Specific Aim#3). 2. Key findings: 2a. We developed and validated custom TMS helmets that allowed us to administer TMS in zero gravity with minimal training and high individual reliability. This TMS helmet method allows for reliable administration of TMS outside the laboratory setting with minimal training and may be useful in patient populations for clinical purposes as well. 2b. In order to consider the use of TMS as a countermeasure for spaceflight the biological efficacy of TMS in a microgravity environment must be determined. Several TMS measures of cortical excitability have been described; however, the most basic is the TMS resting motor threshold (rMT). The rMT is believed to be an indicator of neuronal membrane excitability. Accurate determination of rMT is of utmost importance in both research and clinical studies as it is the unit most commonly used for rTMS dosing. Inaccurate estimation of rMT can lead to subject/patient over- or under-dosing. Here we have shown that rMT levels were 12.6% lower in zero gravity than they were pre- and post-flight in Earth gravity. These rMT changes were transient and did not persist after flight, and were not related to age, gender, or subjective emotional arousal at the time of data acquisition. 3. Our assessment of rMT in the zero gravity condition is an important baseline data point in understanding the response of the brain to acutely altered gravity and will facilitate the design of TMS treatment protocols for use on future spaceflight missions. Our results suggest that clinicians will need to consider altered corticospinal excitability during microgravity when designing TMS protocols. These results could also contribute to the understanding of altered inflight responses reported for psychotropic medications by astronauts and may be useful for physicians and pharmacists when prescribing these medications. 4. Once Covid19 restrictions are lifted, we will continue with enrollment of the dose-finding study (SA#1). Currently 8 of 40 subjects are complete, with 10 more subjects which have been pre-screened and are awaiting enrollment. Once this study is complete and we have identified an optimal TMS dose, we will move immediately into study 2 to evaluate safety and efficacy (SA#2).
Bibliography Type:	Description: (Last Updated: 01/15/2026)
Articles in Peer-reviewed Journals	Badran BW, Caulfield KA, Lopez JL, Cox C, Stomberg-Firestein S, Devries WH, McTeague L, George MS, Roberts DR. "Personalized TMS helmets for quick and reliable TMS administration outside of a laboratory setting." Brain Stimulation. May-Jun 2020;13(3):551-3. https://doi:.org/10.1016/j.brs.2020.01.009 ; Pubmed PMID: 32289675 , May-2020
Articles in Peer-reviewed Journals	Caulfield KA, Lopez JW, Cox, CE, Roberts DR, McTeague LM. "A new, open-source 3D-printed transcranial magnetic stimulation (TMS) coil tracker holder for double blind, sham-controlled neuronavigation studies." Brain Stimulation. May-Jun 2020;13(3):600-2. https://doi.org/10.1016/j.brs.2020.02.003 ; PubMed PMID: 32289684 , May-2020